Quick Start Guide
Active Splitter
ASP-015-50-A



1 About This Guide
This document is a condensed quick-start guide intended to support basic installation and operation of the device. This guide does **not** replace the full product manual. The complete operation manual can be downloaded at: https://flim-labs.github.io/docs/asplitter/Asplitter-manual.pdf

Scan the QR code to open the full manual.

> Reading the complete manual is strongly recommended before operating the device, unless the user is already familiar with similar laboratory instrumentation.

2 Safety Information

> Do not operate the device if it is damaged or if safe operating conditions cannot be ensured.

- Read the full manual before installation and use.
- Operate only within the specified electrical and environmental limits.
- Do not open the enclosure.
- Disconnect power before modifying connections.
- Use only undamaged and properly rated cables.

3 Intended Use
The Active Splitter is intended as a compact electronic module for the distribution and duplication of high-frequency signals by professional users in research laboratories, companies, and universities.

It may be used as a stand-alone unit or integrated into custom measurement systems and instrumentation.

The device must be operated only by qualified personnel familiar with laboratory electronic instrumentation.

4 Product Overview
The Active Splitter is a compact electronic module designed for the controlled duplication and distribution of high-frequency electronic signals. It is primarily used in scientific instrumentation, timing systems, event counting, and fluorescence lifetime imaging microscopy (FLIM) applications, where reliable signal replication without degradation of timing or electrical characteristics is required.

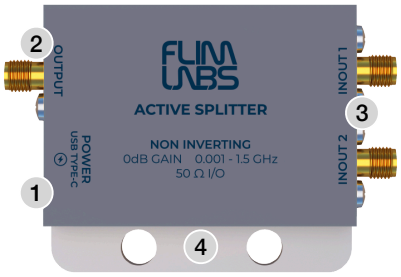
The device features two bidirectional SMA connectors operating in loop-through configuration and one dedicated SMA output. The input signal can be applied to either of the loop-through ports and is forwarded unchanged to the second port. In parallel, the signal is actively replicated and provided at the dedicated output as an impedance-matched copy.

Key Features

- Two SMA loop-through ports
- One dedicated SMA output for the replicated signal
- USB Type-C 5 V power supply
- Impedance-matched signal duplication
- No user configuration required
- Compact laboratory form factor
- Suitable for positive or negative input signals

5 Components Overview
Overview of the key components of the device.

1. USB Type-C power input
2. SMA loop-through port 1
3. SMA loop-through port 2
4. SMA output (active replicated signal)
5. Mounting holes for mechanical fixing



6 Scope of Delivery
- Active Splitter
- USB Type-C power cable
- Quick start guide
- Declaration of conformity

7 Setup
Install the device on a stable laboratory surface. Before installation:

- inspect the device and accessories for visible damage
- verify connectors and cables are intact
- verify that the input signal meets the requirements given in Table 1

8 Connections
Power connection

- Connect the USB Type-C cable to the device power input.
- Connect the other end to a suitable 5 V DC power source such as a computer USB port, a power bank, or a certified 5 V DC adapter.
- Verify visually that the LED power indicator confirms correct device power-up.

Signal connection

- Apply the input signal to either of the two SMA loop-through ports.
- Connect the second loop-through port and the dedicated SMA output to the downstream measurement or processing devices as required.

9 Configuration
After completing all electrical connections, no further configuration is required.

Verify that the device powers up correctly and that stable signals are available on the loop-through and replicated output channels under the expected operating conditions.

10 Operation
Once the power supply and signal connections have been correctly established, the Asplitter operates autonomously without further user intervention.

11 Further Information
Additional information and downloads are available on the product website: https://www.flimlabs.com/products/active-splitter/.

12 Product Label

Active Splitter

P/N: ASP-015-50-A
S/N: ASP-XXXX-XXX
5 V ⎓ 0.1 A | IP40

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13 Technical Specifications

Item	ASP-015-50-A
Electrical Specifications	
Electrical safety class	Class III
Power supply	USB Type-C, 5 V DC
Current consumption (max)	0.1 A
Power consumption (max)	0.5 W
Signal Specifications	
Input/Output connections	2x SMA (loop-through), 1x SMA output
Input polarity	positive or negative
Minimum input pulse width	>1 ns
Maximum input pulse height (abs)	5 V
Impedance	50 Ω
Timing variation	<1 ps
Lower frequency limit	30 kHz
Duplication loss	0 dB
Maximum output amplitude	0.87 V rms
Physical Specifications	
Dimensions (L x W x H)	60 mm x 42 mm x 14 mm
Weight	80 g
Protection rating	IP40
Environmental Specifications	
Operating temperature	0 °C to +40°C
Storage temperature	-20 °C to +70°C
Humidity	up to 80% non-condensing

Table 1: Technical specifications of the Active Splitter.

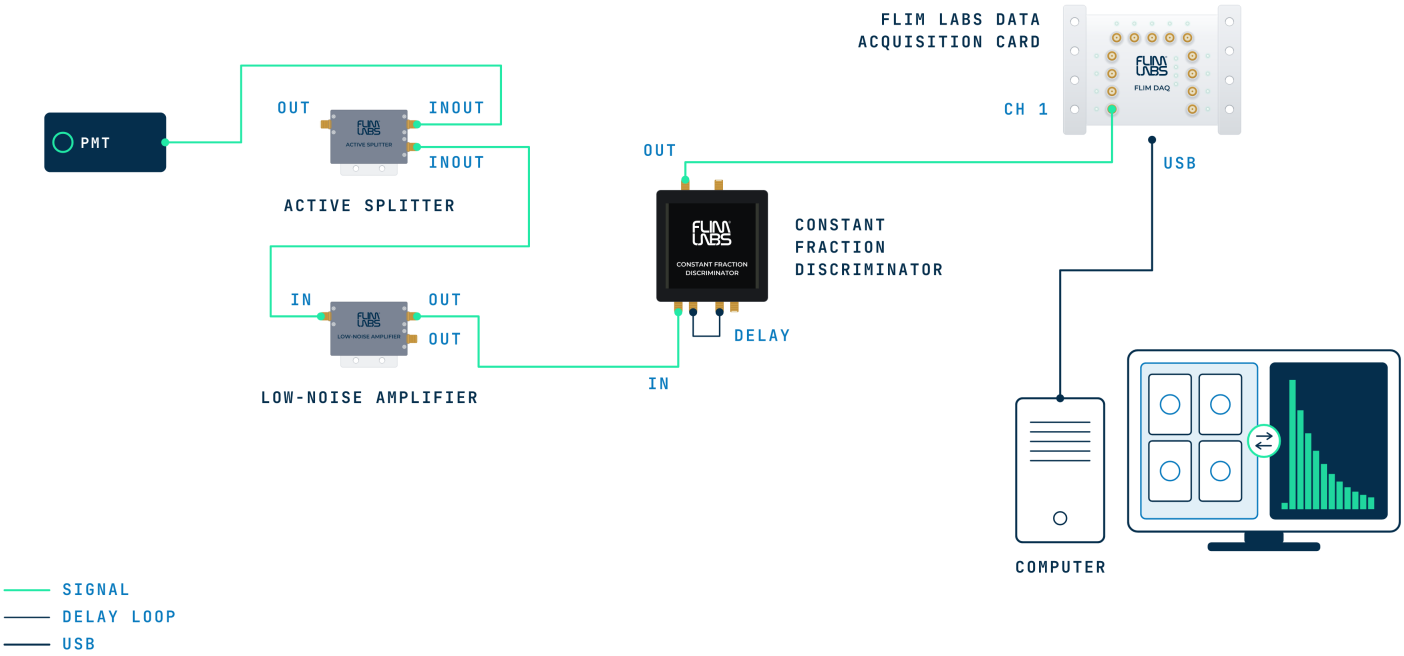


Figure 1: Example configuration for connecting a PMT using the active splitter.